MEASUREMENT UNCERTAINTIES OF THE CALIBRATION OF MULTI-JUNCTION SOLAR CELLS

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ABSTRACT

Measurement uncertainties in the calibration of multi-junction (MJ) cells are investigated. The focus is placed on the uncertainties originating from the main error sources when calibrating solar cells. These are the simulator spectrum and the spectral response measurement. Both uncertainties lead to an increase in the uncertainty of the photocurrents of the subcells. The correlation between the uncertainties of the cell parameters resulting from the above mentioned error sources is addressed. It will be shown that the correlation between the uncertainties of the cell parameters in a MJ cell differs from that in a single-junction (SJ) cell. The spectrometric characterization technique is used in a slightly modified way to reveal the dependencies of the different cell parameters on the uncertainties of the photocurrent in the subcells. For a triple-junction cell an expanded uncertainty of 2.4 % in power with a 95 % confidence was determined.

INTRODUCTION AND MOTIVATION

In several standards and drafts, e.g. [1-4] guidelines for the calibration of solar cells, both single-junction (SJ) and multi-junction (MJ) cells, are given. Standard testing conditions (STC) for different environments have been constituted and correction procedures for taking into account deviations from STCs have been developed, thereby assuring international comparability of measurement procedures and results. Nevertheless, approaches for the calibration of solar cells are manifold. Therefore, the reliability of a measurement strongly depends on the test set-up and the components used in the measurement procedure. Thus, an uncertainty analysis is mandatory for each measurement procedure allowing an assessment of the validity of a measured efficiency. Typically, published uncertainties of solar cell efficiencies are in the range of 2–5 % [5].

Differences in the measurement procedures arise fundamentally from the varied methods of achieving the correct spectral conditions tabulated in the mentioned standards. Some calibration laboratories rely on outdoor measurements under real sunlight, while others calibrate the cells indoors using sun simulators. Both approaches have in common that the spectra under which the cells are measured never match the tabulated ones. Thus, corrections must always be applied to the measurements. It is known from experience that this procedure has the major impact on the overall uncertainty.

In the following the case shall be considered that the cell under test (test cell) is calibrated indoors using a sun simulator against a measurement standard which here is a primary calibrated reference solar cell. Then the major uncertainty originates from uncertainties in the correct adjustment of the sun simulator. Uncertainty in the simulator spectrum originates usually from the following components:

- the measurement standard (usually one or more primary calibrated reference cells)
- the spectral distribution of the spectrum under which the cell is measured
- spectral response of the test cell

The uncertainty in the photocurrent of the test cell is caused, directly or indirectly, by these three factors. These in turn lead to uncertainties of the efficiency. When measuring SJ cells the uncertainty in efficiency is at least as high as that of the photocurrent or the short-circuit current (I_{SC}).

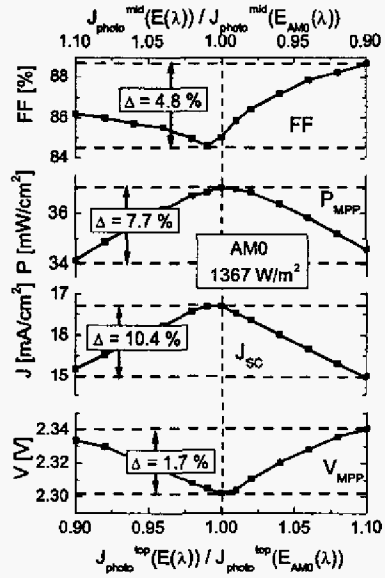

Fig. 1: Spectrometric characterization of a GaInAs/GaInP/Ge 3J cell. The deviation of the photocurrent is partly cancelled out by the FF and V_{MPP}. Thus the deviation of P_{MPP} is less than that of I_{SC}.

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However, this is not necessarily true for MJ cells as can be seen in Fig. 1. Here, the spectrometric characterization [6] of a GaInAs/GaInP/Ge triple-junction (3J) cell is shown. In the course of a spectrometric characterization the photocurrents of top and middle cell are varied in a defined way according to:

$$\frac{J_{photo}^{top}(E(\lambda))}{J_{photo}^{top}(E_{AM0}(\lambda))}+\frac{J_{photo}^{mid}(E(\lambda))}{J_{photo}^{mid}(E_{AM0}(\lambda))}=2 \qquad (1)$$

where J_{photo}^{top} and J_{photo}^{mid} are the photocurrents of top and middle cells and $E(\lambda)$ is the spectral irradiance. The photocurrent of the bottom cell is kept constant at the value under AM0 conditions.

The procedure expressed by Eq. (1) is as follows:

1. Adjust the spectrum in a way that the top cell generates x % less photocurrent compared to the photocurrent under AM0 conditions while the middle cell generates x % more photocurrent as under AM0 conditions.
2. Measure the I-V curve of the cell
3. Repeat steps 1 and 2 by changing x from -0.1 to 0.1 (this is the interval covered in Fig 1).

The cell parameters are then plotted versus photocurrent of the top cell normalized by its photocurrent under the reference spectrum (AM0 in this case).

While the I_{SC} has a maximum when the cell is current matched, other cell parameters such as fill factor (FF) and voltage at the maximum power point (V_{MPP}) have minima at this point. It is clear that part of the decrease in total current of the 3J cell due to current mismatch is cancelled out by the increase of FF and V_{MPP}. This behavior is always recognized when performing a spectrometric characterization of a MJ cell. Thus, this measurement is evidence of the more complex correlations between the cell parameters in MJ cells. However, the way the spectrometric characterization is usually applied, namely according to Eq. (1), covers only a specific situation of uncertainties of the photocurrents without even considering the uncertainties of the photocurrent of the bottom cell.

In the following a detailed uncertainty analysis of the measurement procedure of MJ cells at Fraunhofer ISE is presented. Uncertainties of the photocurrents of the subcells in a triple 3J cell are determined. Using the spectrometric characterization technique in a slightly modified way, by measuring the I-V characteristic of a 3J cell under different spectral conditions, the influence of the uncertainties in photocurrent of the subcells on the uncertainties of the cell parameter of the whole device is investigated.

METHOD

Measurement procedure at ISE CalLab

Fig. 2 shows a sketch of the multi-source sun simulator of ISE CalLab where the MJ (dual- and triple-junction) cells are calibrated. The setup is equipped with three different light sources – a 2.5 kW xenon arc lamp and two halogen tungsten lamp fields - that can be adjusted independently. Thus, a huge variety of spectral distributions can be set. On the right side of the figure an example of the settings for an AM0 measurement is shown. In the upper graph the spectral distributions of the three different light sources are given. In the lower graph one can see the combined output of the three light sources in comparison to the AM0 spectrum. The adjustment of all light sources depends on the spectral distributions of the light sources and the spectral response of the test cell and is determined by the following linear equation system:

$$\sum_{i=1}^{3} A_i \int_{\lambda} s_j(\lambda) e_i(\lambda) d\lambda = \int_{\lambda} s_j(\lambda) E_{ref}(\lambda) d\lambda \quad j = 1, 2, 3 \qquad (2)$$

with s_j being the relative spectral response of subcell j, e_i the relative spectral distribution of light source i, E_{ref} the absolute spectral irradiance of the reference spectrum and A_i the variables determined by the linear equation system. Eq. (2) expresses that subcell j in a MJ device generates the same photocurrent under the simulator spectrum as it would generate under reference conditions. The linear equation system can be solved using e.g. Cramer's rule in order to obtain values for the A_i. Then the different light sources are adjusted by using a reference cell with known absolute spectral response S_{ref} according to:

$$J_{ref,i} = A_i \int_{\lambda} S_{ref}(\lambda) e_i(\lambda) d\lambda \qquad (3)$$

with $J_{ref,i}$ being the photocurrent of the reference cell under light source i.

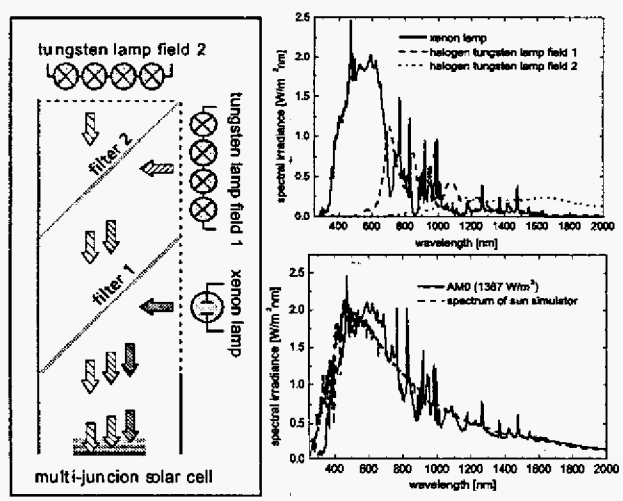

Fig. 2: Left: sketch of the multi-source sun-simulator of ISE CalLab. Right: in the upper graph the spectral distributions of the three light sources are shown. In the lower graph the combined output of all light sources is given compared with the AM0 spectrum.

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Uncertainty analysis

In the following, the uncertainty analysis is performed considering the case of a 3J that is calibrated under AM0 conditions. The uncertainty of the adjustment of the simulator and thus the photocurrents of the subcells has two major sources. The first is from the uncertainties of the factor A_i in Eq. (2). The second is the uncertainties of the input parameters of Eq. (3).

The uncertainties of the factor A_i depends on the uncertainties of the relative courses of spectral response of the test cell and the light sources (see Eq. (2)). A detailed uncertainty analysis for both, spectra and spectral response measurements, was performed. Thereby, it was also considered that some of the uncertainties are wavelength dependent.

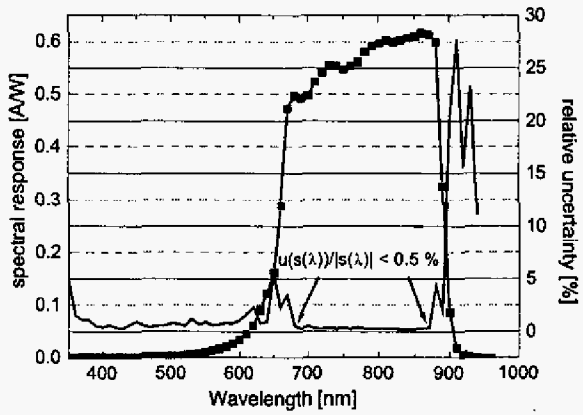

Fig. 3: Spectral response of a GaInAs middle cell and the relative uncertainty. The left axis corresponds to the square symbols line while the right axis corresponds to the plain line. In the range between 680 to 870 nm the relative uncertainty is lower than 0.5 %.

In Fig. 3 as an example the spectral response of a GaInAs middle cell in the 3J device is given together with its relative uncertainty. Since in Eq. (2) only relative values are needed the uncertainty analysis can be confined to the uncertainties affecting only the shape of the spectral response not the absolute values. As can be seen from Fig. 3 the uncertainties can reach very high values of almost 30 %. These high uncertainties occur when the derivative of the spectral response changes rapidly as e.g. near the band edge. In this wavelength range the main contribution to the total uncertainty originates from the resolution of the monochromatic light, which is in this case 5 nm at FWHM. In other regions where the curve is relatively flat the uncertainty is limited by the lock-in accuracy and is usually below 0.5 %. The major contributions to the uncertainties of the three light sources are introduced by the uncertainties originating from the measurement with spectroradiometers. Taking these into account, the uncertainties of the integrals J in Eq. (2) are determined by simulation and by using the following equation:

$$u(J) = \int_{\lambda} u(s(\lambda))e(\lambda)d\lambda + \int_{\lambda} s(\lambda)u(e(\lambda))d\lambda \qquad (4)$$

where $u(s(\lambda))$ and $u(e(\lambda))$ are the wavelength dependent uncertainties of the relative spectral response s and the relative spectral irradiance e.

These uncertainties are then used to calculate the uncertainties of the factor A_i by the following equation in accordance with [7]:

$$u(A_i) = \sqrt{\sum_j \left(\frac{\partial A_i}{\partial J_j}\right)^2 \left(u(J_j)\right)^2} \qquad (5)$$

where J_i is the current density of integral i (cp. Eq. (2)). Each of the factors A_i depends on 12 integrals.

The uncertainties in $J_{ref,i}$ are determined in an analogous way. The $J_{ref,i}$ in turn affect the uncertainties of the simulator spectra e_i by an additional term. Finally, the uncertainties in the photocurrents of the subcells of the 3J cell can be calculated. In the given example the relative uncertainties for the photocurrents of the three subcells are 2.5 % for the top and the middle cell and 5 % for the bottom cell.

The influence of the uncertainties of the subcells on the uncertainty of the whole device is now assessed experimentally. Similar to the spectrometric characterization the spectral conditions are changed and I-V curves are measured. But in this case the spectral conditions are changed in a way that only the photocurrent of one subcell is varied while the others remain constant (thus not in accordance with Eq. (1)).

RESULTS AND DISCUSSION

In Fig. 4 the cell parameters of the 3J cell are plotted versus the variation of the photocurrent of the different subcells cell.

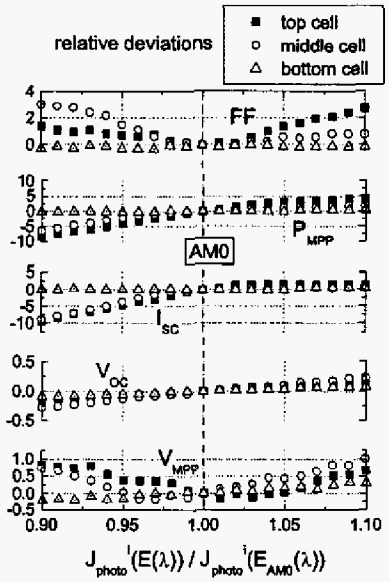

Fig. 4: Cell parameters of the 3J cell versus variation of the photocurrent of the different subcells in each case with the other subcells kept at the values under AM0 conditions.

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Obviously, the photocurrent of the whole device is strongly affected when the subcell becomes the current limiting cell. Then the I_{SC} of the 3J cell changes linearly with the photocurrent of the subcell. Compared to I_{SC} the power is less affected when one of the subcells becomes current limiting. This is the result of the behavior of the FF and the V_{MPP} which partly cancel out the decrease of I_{SC}. On the other hand the power is slightly more affected than the I_{SC} when one of the subcells has excess current.

In order to analyze the uncertainty of the different cell parameters quantitatively, a mean standard deviation for each cell parameter was calculated taking into account the uncertainties of the photocurrents of the corresponding subcell. The mean standard deviations were calculated using the following equation:

$$\overline{u(Y_i)} = \sqrt{\frac{1}{2\Delta}\int_{1-\Delta}^{1+\Delta}(Y_i(x)-Y_i(1))^2\,dx} \qquad (6)$$

where Y_i are the cell parameters, $Y_i(1)$ are the cell parameters measured under "best" AM0 conditions and Δ the relative uncertainty of the photocurrent of the appropriate subcell. The results are given in Table 1. The uncertainties only originating from uncertainties in the photocurrent of the subcell given in the 5th row of Table 1 were calculated in accordance with [7]. In the given case the overall uncertainty is mostly determined by the uncertainties of the top cell. This can differ from cell to cell and depends strongly on which of the subcells is the current limiting cell under reference conditions. As the bottom cell has a lot of excess current the contribution to the overall uncertainty is lower. In any case, it can be stated that the uncertainties in the photocurrent of the subcells do not lead to uncertainties in I_{SC} as high as would be expected in the case of SJ cells. As an important result it can be also stated that at least in the given case the uncertainty in I_{SC} is higher than the uncertainty in P_{MPP}.

	I_{SC}	V_{OC}	P_{MPP}	FF	I_{MPP}	V_{MPP}
$u_{Jtop}(Y_i)/Y_i$ [%]	1.14	0.02	1.09	0.19	1.22	0.15
$u_{Jmid}(Y_i)/Y_i$ [%]	0.52	0.04	0.44	0.14	0.33	0.13
$u_{Jbot}(Y_i)/Y_i$ [%]	0.10	0.03	0.11	0.17	0.13	0.08
$u_{Jges}(Y_i)/Y_i$ [%]	**1.26**	**0.05**	**1.19**	**0.29**	**1.27**	**0.21**
$U(Y_i)/Y_i$ [%]	**2.52**	**0.54**	**2.42**	**0.61**	**2.56**	**0.77**

Table 1: Relative uncertainties of the cell parameters of a 3J device originating from uncertainties of the photocurrents of the three subcells. The 4th row shows the overall uncertainty arising from uncertainties in photocurrents of the subcells. In the last row the expanded uncertainties with 95 % confidence are given.

In order to give the final uncertainties of all cell parameters additional uncertainty sources have to be taken into account, like e.g. temperature. Nevertheless, these additional uncertainties are small compared to the those in the spectrum. The expanded uncertainties with a 95 % confidence (coverage = 2) are given in the last row of Table 1.

SUMMARY

Uncertainties in the photocurrent of the subcells (in this case between 2.5 % and 5 %) do not lead necessarily to such high uncertainties in the whole 3J device. The reason for this is that uncertainties in the photocurrent of the subcells have less effect on the uncertainties in a 3J device when the subcell is not the current limiting cell. It was also shown that uncertainties in P_{MPP} can be lower than uncertainties in I_{SC} due to the complex correlations of the cell parameters in a MJ cell.

Since the uncertainties in a MJ cell depend on the current mismatch of the subcells under reference conditions an uncertainty analysis should be performed for each cell type and each reference condition. The development of a generalized method that can be easily applied to every cell type and measurement condition is an ongoing work at Fraunhofer ISE.

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